

Given the following three simple documents:

Doc 1 = "This is a short sentence"

Doc 2 = "Yet another short sentence"

Doc 3 = "Definitely a long sentence"

Please generate three document vectors (i.e., Doc 1, Doc 2, and Doc 3) for each of the following methods respectively:

- (a) Bag of Words
- (b) Bag of 2-grams
- (c) TF-IDF

(a)	<i>This</i>	<i>Is</i>	<i>A</i>	<i>Short</i>	<i>Sentence</i>	<i>Yet</i>	<i>Another</i>	<i>Definitely</i>	<i>Long</i>
<i>Doc 1</i>	1	1	1	1	1	0	0	0	0
<i>Doc 2</i>	0	0	0	1	1	1	1	0	0
<i>Doc 3</i>	0	0	1	0	1	0	0	1	1

Doc 1 = [1,1,1,1,1,0,0,0,0]

Doc 2 = [0,0,0,1,1,1,1,0,0]

Doc 3 = [0,0,1,0,1,0,0,1,1]

(b)	<i>This-is</i>	<i>Is-a</i>	<i>a-short</i>	<i>Short-sentence</i>	<i>Yet-another</i>	<i>Another-short</i>	<i>Definitely-a</i>	<i>a-long</i>	<i>Long-sentence</i>
<i>Doc 1</i>	1	1	1	1	0	0	0	0	0
<i>Doc 2</i>	0	0	0	1	1	1	0	0	0
<i>Doc 3</i>	0	0	0	0	0	0	1	1	1

Doc 1 = [1,1,1,1,0,0,0,0,0]

Doc 2 = [0,0,0,1,1,1,0,0,0]

Doc 3 = [0,0,0,0,0,0,1,1,1]

(c)	<i>This</i>	<i>Is</i>	<i>A</i>	<i>Short</i>	<i>Sentence</i>	<i>Yet</i>	<i>Another</i>	<i>Definitely</i>	<i>Long</i>
<i>Doc 1</i>	0.095	0.095	0.035	0.035	0	0	0	0	0
<i>Doc 2</i>	0	0	0	0.044	0	0.119	0.119	0	0
<i>Doc 3</i>	0	0	0.044	0	0	0	0	0.119	0.119

Doc 1 = [0.095,0.095,0.035,0.035,0,0,0,0,0]

Doc 2 = [0,0,0,0.044,0,0.119,0.119,0,0]

Doc 3 = [0,0,0.044,0,0,0,0.119,0.119]